Snowpipe

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Section Introduction-Snowpipe

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What is Snowpipe. What use cases does it work best for?

Snowpipe is a continuous data ingestion service designed to automatically load data files into Snowflake tables as soon as the files become available in an external stage (AWS S3).

Snowpipe enables near real-time data loading, allowing Snowflake projects to keep their Snowflake tables up-to-date with minimal latency.

Features of Snowpipe

- Snowpipe is designed for continuous loading of data from cloud storage into Snowflake.
- Snowpipe eliminates the need to schedule jobs to load data to Snowflake.
- Snowpipe eliminates the need to manually execute COPY commands to load data.
- Snowpipe uses s3 event notifications to become aware if new files have arrived in cloud storage and process them.
- Snowpipe is recommended when the files do not have a fixed schedule and can arrive irregularly or in-frequently and need to be immediately processed.
- Snowflake works best for files with a size range of 10 MB and 250 MB.
- Snowflake is serverless which means that one do not provide compute to Snowpipe. It gets the compute needed to load files provisioned automatically by Snowflake.
- Snowpipe supports all file formats supported by the COPY command including semi-structured data.
- Data can be transformed or cast into a different datatype in the COPY when loading data from s3 to Snowflake tables.

What is Snowpipe. What use cases does it work best for?

- Pipe can be paused by using ALTER PIPE <pipename> SET PIPE_EXECUTION_PAUSED=TRUE.
- Pipe status can be checked by SELECT SYSTEM$PIPE_STATUS('<PipeName>').
- INFORMATION_SCHEMA.COPY_HISTORY provides information about the COPY commands run by the pipe.
- INFORMATION_SCHEMA.VALIDATE_PIPE_LOAD provides detailed row level information about errors rows encountered in the pipe Copy if any and is used for troubleshooting pipe loads.
- INFORMATION_SCHEMA.PIPE_USAGE_HISTORY provides information on billing and costs related to Snowpipe.

Troubleshooting Snowpipe errors

All data loading processes will encounter errors eventually and hence Data Engineers need to know how to fix Snowpipe load errors and re-load failed files.

Some of the issues that are encountered are:

- File not being loaded because the pipe is paused.
- File rejects due to data issues in the file/ Data Type mismatch between file and target table.
- Mismatch in number of columns between file and table structure.

Steps to troubleshoot Snowpipe errors:

Step 1 : Check the status of the pipe. SYSTEM$PIPE_STATUS(‘<pipename>’). If paused then use SYSTEM$PIPE_FORCE_RESUME (‘<pipename>’) to resume the pipe.

Step 2: Check for errors in INFORMATION_SCHEMA.COPY_HISTORY to see error messages.

Step 3: Check for errors in INFORMATION_SCHEMA.VALIDATE_PIPE_LOAD to see detailed error messages.

Step 4: Fix the issues causing the failure.

Step 5: Re-process the files using COPY or re-uploading the updated files to cloud storage.

Determine Snowpipe Costs

Snowpipe cost can really add up, especially if there is a high volume of very small files being ingested continuously.

Costs can be monitored by going to INFORMATION_SCHEMA.PIPE_USAGE_HISTORY.

```
SELECT * FROM TABLE(INFORMATION_SCHEMA.PIPE_USAGE_HISTORY(
          DATE_RANGE_START=>TO_TIMESTAMP_TZ('2024-07-03 00:00:00.000 -0700'),
          DATE_RANGE_END    =>TO_TIMESTAMP_TZ('2024-07-19 00:00:00.000 -0700')));
```

```
SELECT *
  FROM TABLE(INFORMATION_SCHEMA.PIPE_USAGE_HISTORY(
    DATE_RANGE_START=>DATEADD('DAY',-14,CURRENT_DATE()),
    DATE_RANGE_END=>CURRENT_DATE()))
```

```
SELECT *
      FROM TABLE(INFORMATION_SCHEMA.PIPE_USAGE_HISTORY(
      DATE_RANGE_START=>DATEADD('HOUR',-12,CURRENT_DATE())))
```

Data is stored in INFORMATION_SCHEMA.PIPE_USAGE_HISTORY for a maximum of 14 days and DATE_RANGE_START must be within the last 14 days.

ACCOUNT_USAGE. PIPE_USAGE_HISTORY stored data for 365 days, however there is a lag of up to 3 hours before data shows up.

Determine Snowpipe Costs

INFORMATION_SCHEMA.PIPE_USAGE_HISTORY returns the following values

Column Name	Data Type	Description
START_TIME	TIMESTAMP_LTZ	Start of the specified time range in which data loads took place.
END_TIME	TIMESTAMP_LTZ	End of the specified time range in which data loads took place.
PIPE_NAME	TEXT	Name of the pipe used for a data load. Displays NULL if no pipe name is specified in the query. Each row includes the totals for all pipes in use within the time range.
CREDITS_USED	TEXT	Number of credits billed for Snowpipe data loads during the START_TIME and END_TIME window.
BYTES_INSERTED	NUMBER	Number of bytes loaded during the START_TIME and END_TIME window.
FILES_INSERTED	NUMBER	Number of files loaded during the START_TIME and END_TIME window.

Limitations of Snowpipe.

- Snowpipe is not recommended for files that are greater than 250 MB.
- Snowpipe is near real time and data will be loaded in a few mins after the file arrives, it is not able to load data in Real time.
- Not all COPY options work with Snowpipe.
    - ON_ERROR = ABORT_STATEMENT.
    - FILES = ('file_name1' [, 'file_name2', ...])
    - FORCE = TRUE | FALSE. ...
    - PURGE = TRUE | FALSE
    - SIZE_LIMIT = num.
    - RETURN_FAILED_ONLY = TRUE | FALSE.
    - VALIDATION_MODE= (RETURN_<n>_ROWS|RETURN_ERRORS|RETURN_ALL_ERRORS)
- Costs of Snowpipe can really add up, especially if there is a high volume of very small files being ingested continuously.
- Snowpipe does not automatically reprocess files that have failed.
- Snowpipe does not load partial data files by default, causing files with a few bad rows to be skipped.

Summary

- Snowpipe is a continuous data ingestion service designed to automatically load data files into Snowflake tables as soon as the files become available in an external stage.
- Snowflake is recommended when the files do not have a fixed schedule and can arrive irregularly or in-frequently and need to be immediately processed.
- Event notifications setup on S3 alert Snowpipe to arrival of new files after which are then ingested.
- Snowpipe works best when files are between 10 MB and 250 MB.
- Snowpipe is serverless which means compute needed to load the files is provided by Snowflake.
- Snowpipe supports all file formats supported by the COPY command and include CSV,JSON, Parquet, ORC and XML data.
- Data can be transformed when loading data with Snowpipe but is not recommended.
- Use INFORMATION_SCHEMA.PIPE_USAGE_HISTORY to check credit consumption of pipes.
- ACCOUNT_USAGE.PIPE_USAGE_HISTORY stores credit consumption for up to 365 days, data latency can be between 45 mins and 3 hours.
- Pipe can be paused by using ALTER PIPE <pipename> SET PIPE_EXECUTION_PAUSED=TRUE.
- Pipe can be resumed by using SYSTEM$PIPE_FORCE_RESUME (‘<pipename>’).
- Pipe status can be checked by using SYSTEM$PIPE_STATUS(‘<pipename>’).
- Use INFORMATION_SCHEMA.COPY_HISTORY to see if the pipe load has errored out.
- Use INFORMATION_SCHEMA.VALIDATE_PIPE_LOAD to see all the errors that are encountered by the pipe.

Thank you

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